



Life history is a key driver of temporal fluctuations in tropical tree abundances

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Edited by Jerome Chave, Centre National de la Recherche Scientifique, Toulouse, France; received November 5, 2024; accepted December 26, 2024 by Editorial Board Member Simon A. Levin

The question of what mechanisms maintain tropical biodiversity is a critical frontier in ecology, intensified by the heightened risk of biodiversity loss faced in tropical regions. Ecological theory has shed light on multiple mechanisms that could lead to the high levels of biodiversity in tropical forests. But variation in species abundances over time may be just as important as overall biodiversity, with a more immediate connection to the risk of extirpation and biodiversity loss. Despite the urgency, our understanding of the primary mechanisms driving fluctuations in species abundances has not been clearly established. Here, we introduce a theoretical framework based around life history; the schedule of birth, growth, and mortality over a lifespan, and its systematic variation across species. We develop a mean field model to predict expected fluctuations in abundance for a focal species in a larger community, and we quantify empirical life history variation among 90 tropical forest species in a 50 ha plot in Panama. Putting theory and data together, we show that life history provides a critical piece of this puzzle, allowing us to explain patterns of abundance fluctuations more accurately than previous models incorporating demographic stochasticity without life history variation, and without introducing unobserved couplings between species and their environment. This framework provides a starting point for more general models that incorporate multiple factors in addition to life history variation, and suggests the potential for a fine-grained assessment of extirpation risk based on the impacts of anthropogenic change on demographic rates across life stages.

biodiversity | fluctuations | ecological theory | life history

This is critical moment in time for changing habitats and climate, with the potential for dramatic future loss of biodiversity. Tropical regions may bear the brunt of this loss (1–4), and to inform our efforts in mitigating these effects, it is key that ecologists develop general and realistic theory for how and when species will coexist (5–11). Statistical tests of biodiversity theory have often focused on static patterns of diversity, such as the overall level of species diversity, or the balance of rare and dominant species in a community (12–16). In the face of these tests, even relatively simple, neutral models of competition have been able to reproduce observations in a number of empirical studies (17, 18). On the other hand, change over time is an inevitability in ecological communities. While early studies suggested that fluctuations in tropical tree abundances were relatively small (8), there has since been an appreciation that abundance fluctuations can accumulate over time, can be broadly spread in magnitude (19), and that tropical species often experience dramatic changes in their abundance on decadal timescales (20). Models of biodiversity have had considerably less success in predicting these dynamics in forest data (21, 22). But if coexistence is to withstand large fluctuations in species abundances, biodiversity theory must be able to predict the size and frequency of potentially impactful events. This reflects an even more general question of how to assess risk, extending across multiple domains (23–27). An important goal is therefore to understand from first principles what determines the size and frequency of population fluctuations, and also what kinds of changes in these fluctuations might be brought about by global (and local) environmental change.

Such changes in populations over time can at least in part be explained by the random element of chance in the processes of birth and death. This is known as demographic noise and is one of the key features of neutral models of biodiversity, where species compete equally for a single resource, like space (6, 17, 28). Demographic noise, in combination with competition for space, predicts population fluctuation sizes that are proportional to the square root of the sum of birth and death rates, and also to the square root of the initial population sizes at the beginning of a time interval. That is, bigger birth and death rates tend to lead to bigger demographic fluctuations, and so do bigger

Significance

Tropical regions face substantial risk of biodiversity loss with global change, and one of the key mediators of this risk is the size of fluctuations in species' abundances over time. Those fluctuations in turn are governed by a complex interplay of interactions, played out through the lens of life history—the schedule of when organisms grow, reproduce, and die. Here, we show that a practical understanding of tropical forest fluctuations can be gained by integrating life history and multispecies competition. We parameterize theory using data from a tropical forest and show that life history, alongside other important factors including environmental variation, is a key driver of abundance fluctuations in this community.

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Author contributions: K.J., J.W.D., and J.P.O. designed research; K.J., J.W.D., and J.P.O. performed research; K.J. and J.P.O. analyzed data; and K.J., J.W.D., and J.P.O. wrote the paper.

The authors declare no competing interest.

This article is a PNAS Direct Submission. J.C. is a guest editor invited by the Editorial Board.

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This article contains supporting information online at <https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2422348122/-/DCSupplemental>.

Published January 24, 2025.

initial population sizes. But when these predictions have been evaluated in tropical forest data, both the overall size of fluctuations predicted by neutral models, and the scaling with initial population size tend to underpredict observed population fluctuations (20).

In addition to demographic noise, natural communities are also subjected to a variety of shifting, extrinsic environmental pressures, arising from sources as varied as drought (29) to disease outbreaks (30). Models incorporating both environmental and demographic noise have provided a potential resolution to the mismatch between ecological theory and observed abundance variation over time in tropical forests (22, 31). This potential resolution relies on two qualitative differences in environmental noise models: First, population fluctuations scale linearly with initial population size (23, 32, 33), and second, environmental noise models tend to introduce free parameters corresponding to the coupling between population and environment, meaning that large proposed amplitudes of environmental noise can lead to correspondingly large population fluctuations (20). On the other hand, it has been challenging to pin down the precise environmental variables driving observed variation across the huge diversity of plant species in tropical forests (31, 34), meaning that it is hard to ground the idea of environmental noise being the primary driver of population fluctuations with specific couplings to environmental variables. For example, while there are species-specific responses to drought (35), broad groups of species are impacted in qualitatively the same way (29). Models incorporating environmental stochasticity thus assume couplings between species and their environment that are in practice difficult to quantify.

Here, we consider an alternative generalization of biodiversity theory, with the goal of identifying whether demographic noise alone can, when suitably combined with life history variation, come much closer to matching observed population fluctuations. We draw inspiration from the known life history differences between tropical trees (36–38), reflected in the way demographic rates vary with life stage (36, 39–43). Prior work has quantified just how life history can serve to amplify or suppress the effects of demographic stochasticity in small communities (44). For example, a pioneer tree with high fecundity and high mortality will more often experience population booms and busts than a slow-growing, low fecundity, shade-tolerant species. But there remain two important knowledge gaps in identifying the combined effects of demographic noise and life history variation on the maintenance of biodiversity. First, to our knowledge there is an absence of theory developed to specifically predict temporal population fluctuations among multiple species with different life history strategies. And second, there has been a relative lack of connection between recent analytical models incorporating life history and competition (44), and data from diverse, natural communities. Since life history differences can be teased out explicitly from empirical data quantifying growth, mortality, and recruitment over time (38, 45–47), there is the potential to quantify the effects of life history differences in combination with demographic noise within communities where these data are available.

Our objective is to identify whether demographic noise alone, when combined with life history variation, can explain observed abundance fluctuations in a tropical forest. Our approach is based around two, linked theoretical models, and the integration of three strands of empirical data to parameterize the modeling. The first model draws from a simulation-based approach, built around life history differences for multiple species, competing for limited

space. We term this model the space-limited matrix community model, or SLMCM, as it is based on matrix population models to characterize life history variation, and reproduction is limited by the availability of space to recruit new individuals. The second model is a coarse-grained version of the SLMCM, and distills the high-dimensionality of this model down to a much smaller set of parameters, one for each species; these are the effective population sizes of each species, and can be thought of as characterizing how much life history amplifies fluctuation sizes over time. The stochastic, mean-field model allows us to make a set of analytical predictions for species abundance fluctuations over time. Whether our combination of models can actually predict abundance fluctuations for tropical species then relies on quantifying life history strategies for many co-occurring species in a diverse community. To assemble the right empirical data for this test, we identify multiple strands of analysis from the Barro Colorado Island 50 ha plot (45), characterizing seedlings, larger trees, and reproductive individuals among 90 species drawn from the plot. Along the way, we ground these findings in the extensive prior work and domain-specific expertise for this tropical community. We use these data to parameterize the SLMCM, which in turn allows us to produce predictions for species abundance fluctuations using the stochastic mean field approach. With those two new ingredients in place, both theoretical and empirical, the comparison of predictions with data is relatively straightforward, and provides clear evidence for the explanatory power of life history in predicting abundance variation over time.

Life History Model and Forest Data

In previous work (44), we introduced a numerical simulation method for competing species with arbitrary life histories: Here, we term this the Space-Limited Matrix Community Model, or SLMCM, and describe it in more detail in *Space-Limited Matrix Community Model*. In this model, species growth and mortality play out across multiple life stages, while recruitment is then limited by the amount of available space. The potential number of recruits for each time step is calculated based on the recruitment of all reproductive individuals. If the number of potential recruits produced, in addition to the extant individuals, exceeds the total available space, new individuals are drawn randomly from the pool of potential recruits so as to make the new total population equal to the system size.

In order to make predictions for fluctuation sizes, the demographic rates of the SLMCM must be parameterized using empirical data. We could in principle then generate numerical predictions for the size and distribution of abundance fluctuations by running many replicates of the SLMCM. Instead, we take a different approach, using a coarse-grained version of this model to make analytical predictions for fluctuation sizes for multiple, competing species. This model is a mean field theory, in the sense that we assume that the focal species interacts with an average taken over other species in the community, and this reduces the complexity of the model to a series of pairwise effects between species (44, 48, 49). Mean field theory has a longstanding precedent in stochastic ecological modeling (50, 51), and its validity rests on the diversity and variability of the local surroundings of each individual (17). The prediction arising from our stochastic mean field model is that fluctuation sizes, Δ_j , for a focal species, labeled by j , depend on its own life history strategy, and those of $S - 1$ competitor species, via the following equation (derived in *Mean Field Model for Fluctuations*):

$$\Delta_j = \langle (p_j(\tau) - p_{j,0})^2 \rangle$$

$$= \tau \frac{p_{j,0}(1 - p_{j,0})}{K} \left(\sigma_j^2(1 - p_{j,0}) + \frac{p_{j,0}}{1 - p_{j,0}} \sum_{i \neq j} \sigma_i^2 p_{i,0} \right). \quad [1]$$

The key connection between this equation and the life history strategy of each species arises through the parameters σ_j^2 , which represent the year-to-year variance in population size experienced by each species, j , and are related to effective population size, N_y , by $\sigma^2 = \frac{K}{N_y}$, where K is the community carrying capacity. While we have one such parameter for each species, this is far less highly parameterized than the SLMCM, which relies on many distinct demographic rates, across multiple life stages for each species, allowing for the relatively simple analytical prediction for Δ_j . On the other hand, the values of σ_j can be derived from the SLMCM, by competing each species j against a neutral copy of itself. We describe this procedure to estimate the σ_j parameters of our theory in *Testing Mean Field Theory*. The other parameters in this equation relate to either community properties, or the details of the specific time window examined; $p_{j,0}$ is the initial frequency of species j among the S total species, and τ is any small time interval, with corrections to this formula entering at order τ^2 .

There are three features of this model that underline the insights gained by integrating population fluctuations and life history. We already alluded to the first when we compared pioneers to shade-tolerant species above; life history differences can impact the amplitude of demographic stochasticity experienced by a population. But here that impact is channeled through the S different parameter values of the variances, σ_j^2 , rather than the much larger set of demographic rates in the SLMCM. Second, fast-growing pioneer species have large values of σ_j^2 , and experience large fluctuations, in contrast to smaller σ_j^2 , and relatively small fluctuations for slow-growing species. This affects fluctuation size Δ_j via the first term in the parenthesis in Eq. 1, proportional to σ_j^2 . Thus, a shade tolerant species (typically with smaller σ_j^2) will tend to experience smaller population fluctuations than its pioneer neighbors. But the coupling of species dynamics through space limitation leads to a new, distinct effect, appearing via the additional sum of terms in the same parenthesis in Eq. 1. The typical size of fluctuations of other species can thus boost or suppress the fluctuations of a focal species, relative to what one would expect given its own life history strategy. Finally, we note that the scaling behavior of Δ_j as a function of initial frequency of the focal species $p_{j,0}$ differs from the scaling behavior in the absence of life history variation. In the standard theory, the sum of all terms in the parenthesis reduces to σ_j^2 , thus leading to a quadratic dependence of fluctuation size on initial frequency proportional to $p_{j,0}(1 - p_{j,0})$. Here, when life history strategies differ across species, there are contributions that arise at cubic order in $p_{j,0}$, and also can depend on the initial frequencies of the other species present.

We have proposed that competition among co-occurring species with different life history strategies is represented by the assumptions of the SLMCM and that the latter is then well-approximated by our stochastic mean field model. Fig. 1 provides explicit evidence for the latter, where the fluctuations of a focal species are amplified due to competition with other species, which have smaller effective population sizes (detailed in *Tests of Mean Field Theory*). The community is simulated using the

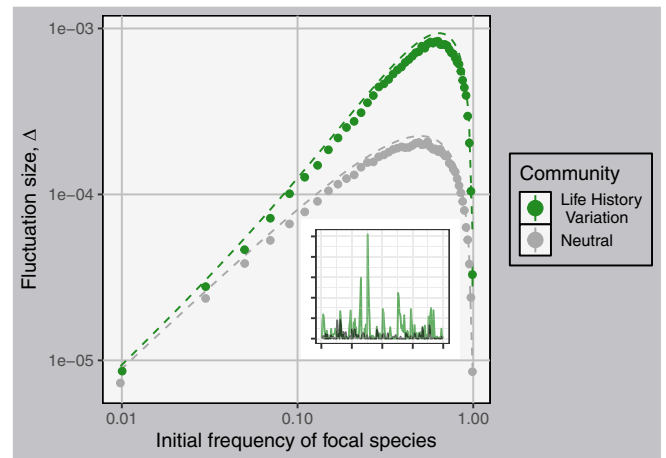


Fig. 1. Theory predicts patterns of fluctuations in interacting communities. We simulated communities based on species with two life stages, as described in *Tests of Mean Field Theory*. Here, we show fluctuation sizes for one focal species, embedded in two distinct community types, distinguished by green and gray dots. Dots represent the binned means of fluctuation size as a function of initial frequency of this species, while error bars represent twice the SE on the mean. In the green community, the focal species competes against nine other species with broadly, but realistically varying life history strategies, as detailed in *Tests of Mean Field Theory*. In the gray community, the focal species competes against nine copies of itself. Dashed green and gray lines then represent the theoretical prediction arising in each case from Eq. 1. The close match between simulated data and the dashed lines demonstrates the validity of our mean field approach, while the clear departure of the data between the two community types demonstrates the importance of life history variation for predicting fluctuation sizes: i.e., the focal species is strongly impacted by both its own life history strategy, as well as the life history strategies of the other species with which it is competing. In this case, those effects systematically increase the typical size of its fluctuations. *Inset* is a time series of fluctuation sizes, demonstrating this difference. In green are 200 timesteps of the community with life history variation, with fluctuation size of the focal species plotted on the y-axis. In black is the set of fluctuation sizes for a neutral community.

SLMCM, and we compare numerical outcomes to the stochastic mean field prediction above. We also show the prediction for what fluctuation sizes *would* be if the focal species were only competing against copies of itself (or equivalently against species with identical life history strategies to its own). The match of the SLMCM and mean field prediction, and the clear departure of both from the neutral community outcomes provide evidence for that validity of our approach, and also show that the fluctuation sizes of one focal species are sensitive to multiple life history strategies integrated across a whole community.

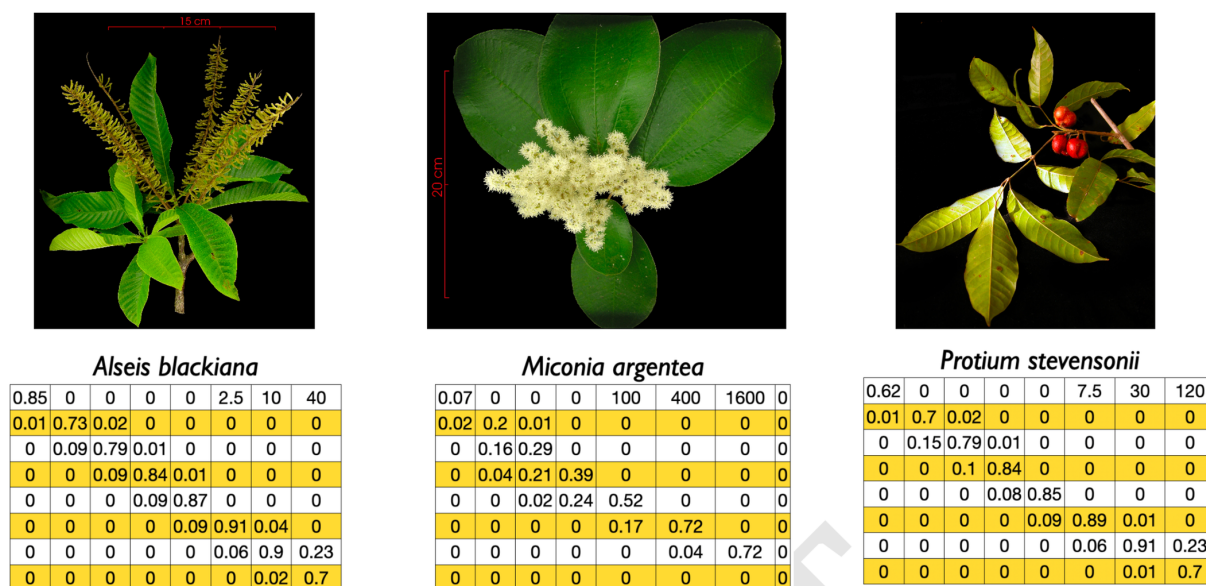
With a model in hand to predict the combined effects of demographic noise and life history differences on observed fluctuations, we now turn to empirical data. The location and species identity of every stem greater than 1cm diameter in the Barro Colorado Island 50 ha plot have been recorded at intervals of five years since 1980 (45). In addition to these census data for 323 distinct species, data collected for seedlings at diameters smaller than 1cm have further enriched our understanding of tree population dynamics across life stages (46). Early analyses of the plot census data suggested a substantial role of chance in the locations of new recruits between species (54), and these data have energized a continuing debate over whether tropical forest tree species depend critically on different environmental conditions or nutrient requirements, or largely compete within the same ecological niche (17, 55–58). Multiple independent studies since have provided evidence for a relatively small number of distinct niches in the 50 ha plot, with substantial overlap and large groups of species competing for similar resources (17, 59–62). Thus,

both demographic stochasticity and life history variation have the potential for key roles in shaping Barro Colorado Island species abundance changes over time.

We characterized life history differences between species at Barro Colorado Island using matrix population models (MPMs), which encode demographic rate changes as a function of an individual's life stage (63, 64). To arrive at these inferred MPMs, we integrated census and seedling data to calculate survival and growth estimates for each of the species modeled in this study (45, 46), as described in *Inference of Barro Colorado Island Matrix Population Models*. We used diameter at breast height (DBH) as a metric to sort populations into eight size classes and encoded the probability of observed individuals moving between these size classes throughout their life cycles. Some species were excluded due to sparsity of observations in one or more classes, leaving the 90 focal species we present here (see *Inference of Barro Colorado Island Matrix Population Models*, for further information on filtering data and constructing the models). Though we do not capture the full species diversity of the plot, these 90 species represent over 90% of all observed stems. Thus, we are able to include the vast majority of individual trees in our downstream analyses. Pairs of censuses from Barro Colorado Island were used to calculate vital rates across size classes across 8, 5-y sampling periods (45, 46). An additional dataset on minimum reproductive diameters for our focal species was used to determine the threshold at which individual trees start to produce seeds ([SI Appendix, section 3](#)). We then assumed that recruitment scales as a power law of diameter to infer relative recruitment rates of each species across the size classes equal to or larger than this minimum reproductive diameter (65, 66). We initially consider recruitment scaling as diameter squared, and generalize this later on.

Results

As a second test, we extend our analysis from the synthetic matrices we tested in Fig. 1, to the empirical life history strategies we inferred for these 90 Barro Colorado Island species. We simulated



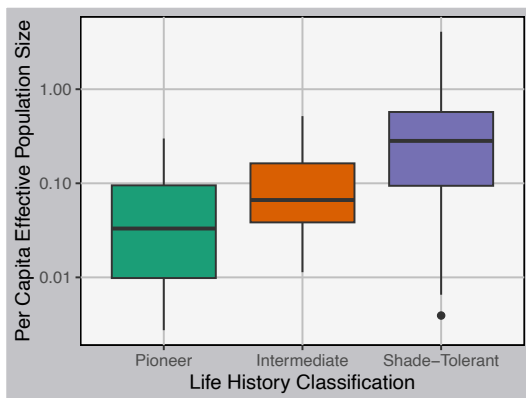


Fig. 3. Effective population size correlates with plant functional types. In prior work (53), tree species in the Barro Colorado Island plot have been classified into multiple functional types. Here, we show a comparison of those types with the continuous variable, effective population size, that characterizes the relative size of fluctuations experienced by different species. 49 of our species were classified in the shade-tolerant class described in ref. 53, 25 as intermediate, and 16 as pioneer species. The *Lower* and *Upper* box hinges correspond to the first and third quartiles, while whiskers extend from each hinge to the largest value, so long as it is no further than 1.5 times the interquartile range away from the hinge. Outliers beyond this are plotted as points. We were unable to reject the null hypothesis that each category of effective population sizes was log-normally distributed, and a one-way ANOVA rejects the hypothesis that all three types (pioneer, intermediate, and shade-tolerant) have the same log mean effective population size, with P -value $2.9 \cdot 10^{-6}$. Individual, two-tailed t tests between the three pairings of functional types (P-I, I-S, and P-S) indicate that all three types have significantly different log means, with P -values of 0.02, 0.0014, and $3.66 \cdot 10^{-5}$, respectively. This demonstrates that effective population size can be grounded by the existing “fast-slow” classification used to distinguish tree species, while also translating this classification into a quantitative variable that can be used to predict abundance variation over time.

competition between these species using our space-limited matrix community model, with a system size comparable to that of the forest dynamics plot, for five timesteps, over multiple simulation runs, as described in *SI Appendix*, section 5. In *SI Appendix*, Fig. 3, we demonstrate the agreement of the mean value of fluctuation size, Δ_j , averaged over these simulations for each species, with the predictions of Eq. 1. Finally, we consider the robustness of inferred effective population size to our assumption of allometric scaling. In addition to the theoretical scaling of recruitment with tree diameter proportional to diameter squared, we explored two alternative scaling exponents: recruitment linear in diameter and recruitment proportional to diameter raised to the fourth power. Collectively, this range of scaling exponents encompasses the broad range of empirical scalings seen across global forest data (65, 66, 69). We found that effective population sizes were largely insensitive to this choice, with Pearson correlations between inferred effective population size ranging from 96.3% to 99.6% among the three choices of recruitment scaling (*SI Appendix*, section 4). Certainly, recruitment scaling is an extremely important demographic parameter. But for our data, we conclude that minimum reproductive diameter, mortality, and growth rate have the strongest influence on effective population size, and thus on our theoretical predictions for abundance variation over time.

Having inferred matrix population models for 90 of the species present on Barro Colorado Island, and tested the robustness of these matrices to our assumptions, we now assess whether these life histories differences impact predictions for species abundance variation over time. To characterize abundance fluctuations, we use the plot census data for all stems >1cm in diameter,

and a time interval of five years. In Fig. 4, we compare the predictions for abundance variation arising from a single-stage model, based on observed rates averaged over all individuals for a given species irrespective of their diameter, to predictions based on our inferred multistage model. The former analyses have been cited to demonstrate the inadequacy of demographic noise alone in explaining observed fluctuations (20, 22, 31), and could be thought of as assuming no complex life history, with every individual within a species treated as at the same developmental stage. This first model does allow for mortality rate to vary across species, and hence is slightly more general than a purely neutral model, but since the demographic rates in this single-stage model simply do not vary much across species, the single-stage model performs poorly, underpredicting fluctuation sizes for the majority of Barro Colorado Island species.

The second set of analyses, shown in the *Right-hand* panels of Fig. 4, is based on our eight-stage MPMs for each species in combination with the mean field model in Eq. 1. We note two outcomes. First, there is a clear difference in predicted abundance variation between models with and without complex life history; so life history does matter for predicting fluctuations. Second, observed fluctuations are now much closer to the predictions of the life history models, shown in the *Right-hand* panels of Fig. 4, while they deviate strongly (as previously observed) from the single-stage model.

Alternative models have been used in previous work to provide an explanation of the fluctuation sizes we show in Fig. 4. These have generally been based around environmental stochasticity, whereby recruitment and mortality parameters themselves fluctuate over time, caused by an assumed coupling between these parameters and a fluctuating environment. The characteristic pattern generated by environmental stochasticity is a power law dependence for fluctuation sizes, Δ , on initial relative abundance. But on a species-by-species basis, very few species cover a broad enough range of initial relative abundance to identify this scaling definitively. Instead, while all species are assumed to be independently affected by the environment, if they have the same overall amplitude and type of both demographic and environmental stochasticity, then a power law scaling fits observed data better than demographic stochasticity alone (20, 22, 31). We describe the parameterization of this power law in *Environmental Noise Model*, and show that the two free parameters of this model can be fitted using observed data. It is challenging to make a direct model comparison, and our goal in this paper is not to prove that environmental noise plays no role in determining fluctuation sizes at all, but rather to show that demographic noise, when coupled to life history variation, can make comparably accurate predictions for the data shown in Fig. 4. With this goal in mind, we find that our model using quadratic allometric scaling of recruitment with diameter makes predictions for abundance fluctuations that are closer to observed fluctuations than the power law model for environmental noise in 42% of observations. A more complete model comparison will require an expression for the likelihood of observations under our life history model, which could readily be compared with the likelihood (or AIC) of the power law model above, or any other model of environmental noise. We note that for our alternative choices of recruitment scaling exponent (described in *Transition Rate and Recruitment Rate Estimation*), this figure was robust, with similar performance (*Environmental Noise Model*). In summary, our model outperforms the environmental stochasticity model in close to half of the observations, despite having no free parameters to tune.

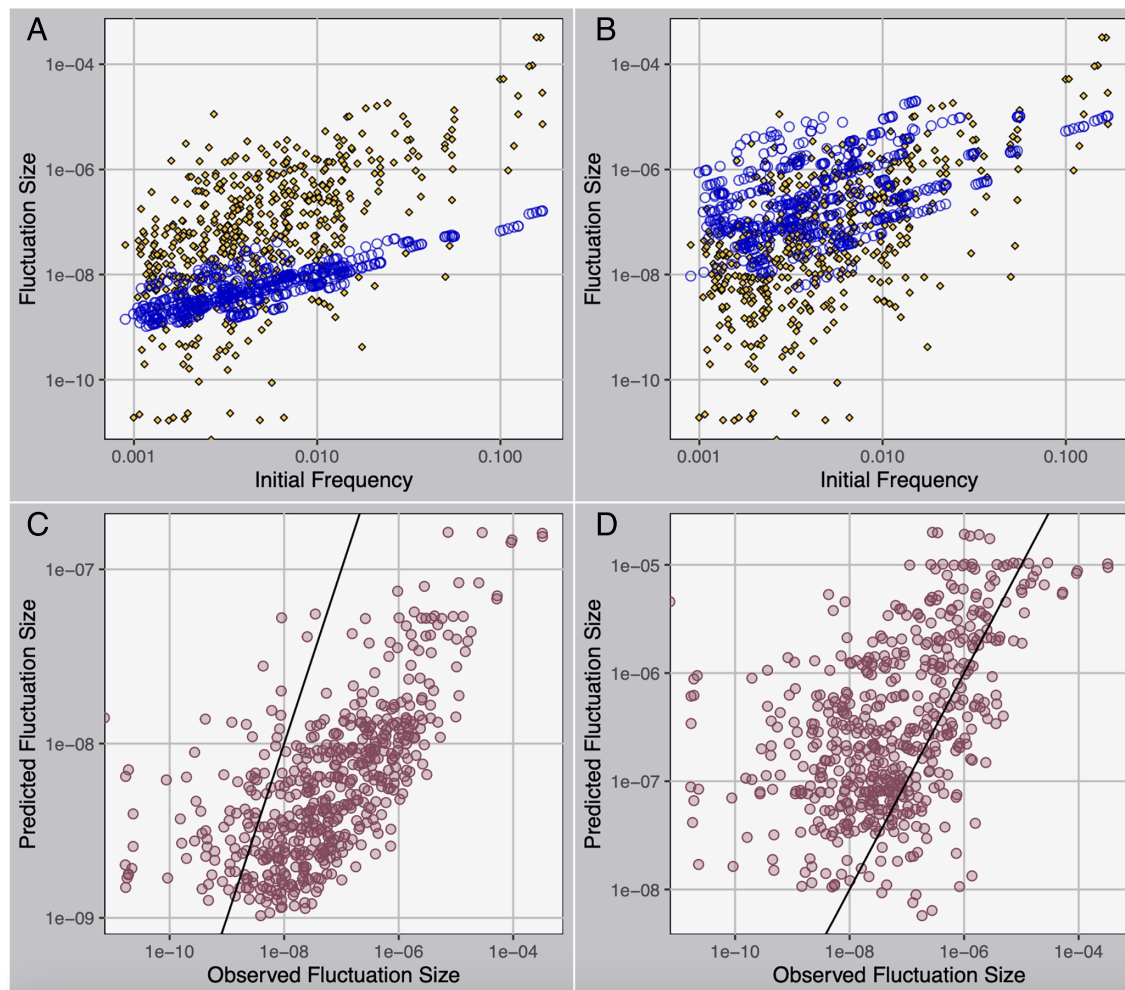


Fig. 4. The combination of demographic noise and life history variation captures patterns of fluctuations. The Top row of panels (A, B) plots predictions and observations for fluctuation sizes as a function of initial frequency for each of 90 species across multiple censuses, while Lower panels (C, D) directly compare how well predictions match observations—the same data plotted in a different way. The Left-hand panels are based on predictions arising for a model with demographic noise and a single life stage, while the Right-hand panels show the impact of adding complex, multi-stage life history. The predicted fluctuations in (A and C) are based on expected fluctuation size, Δ , as defined in Eq. 1, for a model where each species is assumed to have a single life stage. The Upper Left panel shows the predictions (blue open circles) overlaid on observations (gold dots), as a function of initial frequency, while the Lower Left panel plots predictions versus observations for each species and pair of censuses, and the black diagonal line indicates a 1 : 1 relationship). The Right hand panels plot the same empirical data, but with predictions now based on our eight-stage matrix population models and mean field theory, described in *Inference of Barro Colorado Island Matrix Population Models*, and Eq. 1, respectively. The resulting predictions largely center around observed fluctuations, with some remaining outliers among the rarer species.

Discussion

Our study is in the context of considerable and longstanding development of theoretical models to explain patterns of biodiversity, and the potential for biodiversity loss (70–73). An earlier focus on static patterns of biodiversity has since been complemented by a realization that variation over time in species abundances may provide a more pertinent set of predictions in the face of anthropogenic change (20, 22, 31, 74–77). Tropical forests are at the nexus of both issues: Extensive census data provide the means to stringently evaluate theoretical predictions, while tropical communities are at extreme risk of substantial biodiversity loss (1–4), underlining the importance of getting those theoretical predictions right.

We introduced a theoretical model to account for the interaction of distinct life history strategies among species competing for limited space, with the goal of understanding how much this interaction impacts the way species abundances vary over time. Our theoretical analysis indicates both that the mean field

approximation of this life history model provides an accurate description of abundance variation in simulated versions of the same theory and also that life history makes an important difference to the net size of this abundance variation relative to models lacking life history. We applied this model to analyze changes in relative species abundance in census data taken from the Barro Colorado Island plot, in combination with the inference of different life history strategies for 90 tree species. The resulting predictions show that life history alone can explain much of the structure of abundance variation, owing to the systematic variation in life history strategies that we uncovered. Our model combining life history and stochastic competition also performed comparably with a model incorporating extrinsic, environmental stochasticity. Environmental noise has previously provided the primary alternative to demographic noise as an explanation of abundance variation over time, but at the expense of introducing additional free parameters.

There are multiple additional processes that could serve to complement our current analysis. First, we did not quantify

the effect of immigration from outside the plot, and thus our model was based around endogenous recruitment from within. Immigration can contribute to changes in species abundances over time, but prior work has estimated that around 7% of recruits arise from immigration events from outside the Barro Colorado Island plot (78, 79), and so we hypothesize that the bulk of the variation in species abundances in these data can be explained using data from within the plot. Still, dispersal-assembly likely plays a key role in maintaining diversity over longer timescales (80). One of the enduring lessons of neutral models is that species co-occurrence may sometimes be better explained as long-term persistence (81), rather than indefinite coexistence (82). This long term co-occurrence may be a balance between local extirpation dynamics, mediated by life history variation, and the introduction of novelty by dispersal or (on long timescales) evolutionary processes. Dispersal might also shed further light on the variation we observe in effective population sizes. Theoretical predictions (44) suggest that persistence time for competing species is maximized when effective population size is inversely proportional to relative abundance, which would lead to a broad spread of effective population sizes in a community like Barro Colorado Island. But a full integration of dispersal from outside, and life history differences within the plot, will be needed to predict longer-term dynamics, alongside classic patterns like the relative species abundance. The accuracy of our mean field modeling approach suggests that incorporating dispersal, and then solving the mean field model over longer timescales will be a fruitful future step in understanding biodiversity maintenance in the presence of life history variation.

Extending beyond the effects of dispersal and demographic noise, in any tropical community we are likely observing the emergent effect of multiple processes. Niche and fitness differences could allow for longer coexistence between species with different effective population sizes (44), and some of these effects could complement our findings that fluctuation sizes vary across species. For example, the idea that rare species have more stable abundances has also been identified in the context of plant-soil feedbacks, where both greenhouse and field studies indicate that rare species are subject to stronger density-dependent effects, dampening abundance fluctuations (83–85). In general, the interplay between life history, dispersal, niche-differentiation, and fitness differences could all contribute to understanding abundance variation over time. Life history itself could also be more complex than our matrix population models; semelparity and heterogeneous seed production across individuals are both relevant in tropical forests (86), while eight of our 90 species form seedbanks (87), which may shift the predicted dynamics of this subset of species. And the difference in space use with life stage could also play an important role.

Finally, while we have shown that demographic noise, passed through the lens of life history, can explain a substantial amount of observed abundance variation, it is inevitable that environmental stochasticity couples to demography. The known differential effects of drought across broad groups of species provide an important example (29, 35). It is likely that these effects will shift our predictions even closer to observed population fluctuation sizes, and indeed, it may be that for some species, those more tightly coupled to environmental variables, that this effect is substantial—while perhaps for others environmental stochasticity will be less important. If these precise environmental couplings are identifiable, that would also lead to the potential to predict directionality, as well as overall size of population fluctuations. Thus, combining both life history and

responses to environmental fluctuations will potentially lead to an important synthesis, extending our understanding of biodiversity maintenance and extinction risk in tropical plant communities.

Life history modeling has played an important role in conservation and management, and the formalization of matrix modeling was largely motivated by the benefits of fine-graining the demographics of vulnerable species (63, 88). We adapt this to our community framework by comparing a demographic noise model of birth and death to our life history model. Focusing on our least abundant species, *Triplaris cumingiana*, we project times to local extinction using both models. The results, elaborated and plotted in *SI Appendix, Fig. 4*, demonstrate that life history can significantly alter management priorities in a community context. The mean times to extinction differ slightly between the two models, but the variance in these times is several times greater for the life history model. Across 100 simulated projections, the demographic noise model never resulted in a local extinction in under 100 y, whereas the life history model did so 17 times. These models did not incorporate stabilizing processes such as niche preferences, or immigration from outside the plot that could buffer local populations, but are presented to show how projections can vary significantly when stage structure is considered. Future work in this direction may need to be simulation-based and to take into account these additional processes, alongside the subtleties in the changing context of which other species are present. This type of longer timescale prediction could be extended to address other areas where neutral models (without life history) have tended to inaccurate poor predictions. For example, neutral models tend to overpredict species ages (81, 89), and mischaracterize phylogenetic tree structure (90). But extending our model to make precise, long-term predictions would require a careful treatment of the context dependency of community dynamics, and also the potential for the evolution of life history strategies over long timeframes (91).

We highlight these examples to demonstrate that, just as a matrix population model can reveal intricacies in a species' demographics, our model provides predictions on the robustness of communities. On the other hand, these predictions do rely on the ability to parameterize the demography of a broad range of species. How best to extend these types of predictions beyond Barro Colorado Island represents an important open question, given that very few plant communities are as well studied as the Barro Colorado Island forest dynamics plot, with multiple censuses and data across a broad range of life stages. Alternatives might include trait-based predictions of demographic rates, or of effective population size. These relationships have a grounding in previous studies on Barro Colorado Island (92, 93), but the generality of how well traits predict demography has been disputed (94, 95). Nevertheless, identifying ways to extend our analyses across a broad range of plant communities is a priority for future work.

These results thus lay the groundwork for extensive future theoretical work, and also have very applied, tangible implications. The analyses described above show that it is not merely a curiosity to observe that changes in demography with life stage lead to a more complete description of species abundance fluctuations. Global change is expected to induce not just changes in plant demography, but changes that may be highly specific to life stage (96–98). Drought tends to impact mortality of large trees much more severely than smaller trees, across species in tropical systems (29, 99) and beyond (100, 101), while defaunation in many tropical forests has led to recruitment-specific impacts for those species that depend on seed dispersal by animals (102).

More generally, species-specific and stage-specific rate changes could dramatically impact community life history variation, and potentially amplify species abundance fluctuations. The key advance of our framework is to underline the connection between demography and fluctuations not just for individual species, but for species in the context of large, diverse communities.

Materials and Methods

Matrix Population Models.

Matrix population models without space limitation. The primary metric of a species' life history strategy in this study is the set of demographic indicators, broken down by ages or stages, as seen in a MPM. These indicators represent survival and recruitment at each age or stage as well as the probability of transition between two stages. Here, we will model and infer stage-structured MPMs. This matrix relates a vector of abundances across life stages from one time step to the next:

$$n_{t+1} = \begin{pmatrix} s_1 & t_{12} + f_2 & \dots \\ t_{21} & s_2 & \dots \\ t_{31} & t_{32} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix} * \begin{pmatrix} n_1 \\ n_2 \\ n_3 \\ \vdots \end{pmatrix} \quad [2]$$

The recruitment terms, denoted by f_i , represent the mean number of offspring produced by an individual of stage i , the growth terms t_{ji} are the probability of an individual in stage i transitioning to stage j , and the survival terms s_i are the probability of a stage i individual remaining in that stage over a given time step (63, 64). The matrix, generally denoted as A , can also be partitioned into components from transitions and recruitment where $A = F + U$. F is a matrix of recruitment terms, and typically consists of entries along its top row, corresponding to the different recruitment rates across different life stages, all resulting in new individuals in the first life stage. U characterizes the survival (diagonal) and transition (off-diagonal) terms as organisms move through different stages during their lifespan. For a stage-dependent mortality rate over 1 time step, one can subtract the column sums of U from 1 to define vector $\mu = 1 - \text{colSums}(A)$. The solution to the vector equation:

$$A * v^0 = \lambda v^0 \quad [3]$$

normalized such that $|v^0| = 1$, then gives a vector of steady proportions between stages. This vector has an associated largest eigenvalue λ which takes a real value and describes a long-term exponential growth rate for a population in steady proportion (63).

The space-limited matrix community model. In ref. 44, we introduced a model to simulate the interactions of multiple species with different life history strategies competing over a common resource or set of resources. Here, we think of that resource as being space limitation, and we call this model the Space-Limited Matrix Community Model, or SLMCM. Our predictions for species abundance fluctuations depends on a parameter that we will calculate using the SLMCM, and we describe its construction here. In essence, this is a model where growth and mortality play out according to a Matrix Population Model without space limitation, as described just above. But recruitment is limited to the amount of available space. Specifically, the demographic parameters associated with each species are given by an empirically measured or synthetically constructed matrix population model. At each time step, transitions between stages and deaths are then tallied. Each individual has transition probabilities as defined in the relevant MPM, and after the number of individuals in each stage following growth and mortality is determined, the number of recruits for this time step is calculated. Individuals in reproductive stage i have an associated f_i from the MPM, which determines the mean number of offspring produced in a time step. Each individual in stage i then produces a number of *potential* recruits in life stage 1, given by a Poisson distribution with mean f_i . The key effect of space limitation in the SLMCM is that the number of realized recruits may be different from this number of potential recruits. If the number of potential recruits produced exceeds the cap on total individuals in the system, the recruits are tallied and then drawn randomly so as to make the new total population

equal to the capped system size. In summary, competition in this model is imposed via space limitation, and this plays out through recruitment limitation, rather than changes in growth or mortality rate that depend on resource or space availability.

For populations where the leading eigenvalue, λ , is close to one, as simulated in ref. 44, the primary determinants of persistence time for a pair of species, as well as of expected fluctuation sizes over short time-frames, are their effective population sizes. These are organisms where the combination of organismal traits and environmental context are such that the population would be slow-growing even in the absence of space limitation and might be reasonably termed stress-tolerators in Grime's three-fold classification (103, 104). We now extend this analysis for populations in a resource-rich environment, where we might expect λ substantially larger than 1—closer to a competitor scenario in Grime's classification (103, 104). We will show that large, equal values λ can lead to induced, effective growth rate differences between species, but that one additional constraint again leads to effective population size being the primary determinant of competitive outcomes.

To show this, we model space limitation in terms of an effect on the realized recruitment, which suggests that for $\lambda \gg 1$, the number of individuals in the first, juvenile life-stage might depart from the leading eigenvector solving Eq. 3. In turn, that departure might impact the effective, deterministic growth rate of each species, taking it away from λ . We can quantify that departure and its impact on the outcome of pairwise competition, initially for a pair of two-stage populations with the same growth rate, λ (which we go on to simulate in *Tests of Mean Field Theory*), and then generalizing to multistage populations (*SI Appendix, section 1*). We parameterize the MPMs for two populations as:

$$\begin{pmatrix} s_1 & f \\ t & s_2 \end{pmatrix} \quad [4]$$

and

$$\begin{pmatrix} S_1 & F \\ T & S_2 \end{pmatrix}. \quad [5]$$

i.e., lower-case symbols correspond to this specific case of Eq. 2 for species 1, while the upper-case symbols correspond to species 2. We denote the leading eigenvectors by $\begin{pmatrix} w \\ 1 \end{pmatrix}$ and $\begin{pmatrix} W \\ 1 \end{pmatrix}$, and in the case of species 1, w satisfies:

$$\begin{aligned} s_1 w + f &= \lambda w, \\ tw + s_2 &= \lambda. \end{aligned} \quad [6]$$

for leading eigenvalue, λ .

We now consider a case where species 1 is the resident species, and the system size is limited to K . Since $\lambda > 1$, the population will approach the system size, and at that point space limitation will stop the population from growing, with a corresponding constraint on the proportions of juveniles to adults. Parameterizing this new ratio as $\begin{pmatrix} w' \\ 1 \end{pmatrix}$, the number of surviving adults and maturing juveniles must reach a steady state value, so that:

$$tw' + s_2 = 1. \quad [7]$$

and hence $w' = \frac{1-s_2}{t}$. We now consider the effect of species 2 growing from rare in this monodominant background. The dynamics of species 2 depend both on its own intrinsic demographic rates, but also on the number of gaps opened at each time step due to mortality, and the total number of juveniles produced across both species during each time step. Since species 1 is initially dominant, both of the latter quantities are dominated by events from species 1. Assuming that species 2 quickly approaches steady state proportions $\begin{pmatrix} W' \\ 1 \end{pmatrix}$, the resulting dynamics of species 2 are then:

$$\begin{aligned} S_1 W' + \frac{F}{f} ((1-s_1-t)w' + (1-s_2)) &= \Gamma W', \\ TW' + S_2 &= \Gamma, \end{aligned} \quad [8]$$

where Γ is an effective year-on-year growth rate to be determined, along with the ratio of juveniles to adults, W' . Simplifying using Eq. 7,

$$S_1 W' + \frac{F}{ft} (1 - s_1)(1 - s_2) = \Gamma W',$$

$$TW' + S_2 = \Gamma. \quad [9]$$

From this analysis, in this background, species 2 dynamics satisfy an effective MPM with matrix:

$$\begin{pmatrix} S_1 & \frac{F}{ft} ((1 - s_1)(1 - s_2)) \\ T & S_2 \end{pmatrix} \quad [10]$$

Growth rate of species 2 is therefore given by the solution with largest real part of the characteristic polynomial for this matrix,

$$\Gamma^2 - (S_1 + S_2)\Gamma + S_1 S_2 - \frac{FT}{ft} (1 - s_1)(1 - s_2) = 0 \quad [11]$$

A minor adaptation of the Routh-Hurwitz theorem (105) tells us that Γ will have at least one solution with real part > 1 , i.e. allowing species 2 to invade, if either $S_1 + S_2 > 2$, or $(1 - s_1)(1 - s_2) < \frac{FT}{ft} (1 - s_1)(1 - s_2)$. Since both survival rates S_1 and S_2 must be less than 1, the first criterion cannot hold. But the latter can be satisfied if:

$$\frac{(1 - S_1)(1 - S_2)}{FT} < \frac{(1 - s_1)(1 - s_2)}{ft}. \quad [12]$$

In this case, species 2 will invade with an initial positive growth rate given by the leading eigenvalue Γ .

We make two additional comments. First, only one of these two species can invade in the background of the other. Either $\frac{(1 - S_1)(1 - S_2)}{FT}$ is less than the corresponding combination of quantities for species 1, in which case species 2 invades species 1, or vice versa, in which species 1 can invade species 2. Second, if we satisfy the constraint

$$\frac{(1 - S_1)(1 - S_2)}{FT} = \frac{(1 - s_1)(1 - s_2)}{ft}, \quad [13]$$

there will be no deterministic growth of either species in the background of the other. In this case, persistence times and fluctuation sizes will again be determined by the effective population sizes of the two species. In [SI Appendix, section 1](#), we generalize this result to populations with arbitrary numbers of life stages.

Mean Field Model for Fluctuations. In this section, we introduce the fundamental theoretical components describing competition between two populations with different life history strategies, and then generalize this to a mean field theory appropriate for multiple populations competing, which will allow us to model fluctuations in a tropical plant community. The relative frequency of two populations experiencing different levels of demographic noise can be modeled using a Kolmogorov forward equation (44, 48). This difference in the amplitude of demographic noise can arise for populations of two distinct species when they use distinct life history strategies and is characterized by variances in year-to-year changes in the respective populations, σ_1^2 and σ_2^2 , which depend on the specifics of the demographic rates for these two species. The precise values of the σ_j^2 parameters can be derived from our SLMCM above, with the procedure described in detail below in *Tests of Mean Field Theory*. In short, though a larger value of σ^2 would arise from an age-structured model with large recruitment, but low survival rates. In this model for two competing species, population sizes fluctuate independently over time, but with the total sum of individuals periodically "reset" to equal the carrying capacity of system, K (49). σ_i can also be expressed in terms of yearly effective population size N_y as $\sigma_i^2 = \frac{K}{N_y}$.

Under these assumptions, the probability $P(p, t)$ that species 1 has frequency p at time t satisfies:

$$\frac{dP}{dt} = -\frac{\partial}{\partial p} (M(p)P) + \frac{1}{2} \frac{\partial^2}{\partial p^2} (V(p)P), \quad [14]$$

where the drift and diffusion coefficients $M(p)$ and $V(p)$ are given by:

$$M(p) = \frac{p(1 - p)}{K} (\sigma_2^2 - \sigma_1^2) \quad [15]$$

$$V(p) = \frac{p(1 - p)}{K} (\sigma_2^2 p + \sigma_1^2 (1 - p)). \quad [16]$$

Assuming short timescales τ , and an initial frequency $p_{1,0}$ for species 1, we can approximate the integration of this forward equation over a time interval τ by truncating the solution at lowest order in an expansion of powers of τ . This results in the following measure of expected fluctuation size, to linear order in τ :

$$\Delta = \langle (p(\tau) - p_{1,0})^2 \rangle = \tau \frac{p_{1,0}(1 - p_{1,0})}{K} (\sigma_2^2 p_{1,0} + \sigma_1^2 (1 - p_{1,0})) \quad [17]$$

with any deterministic effects arising from $M(p)$ entering at $O(\tau^2)$.

We next introduce a mean field approximation for a multispecies system, where each of S total species, labeled by i , potentially has its own distinct life history strategy and corresponding variance, σ_i^2 . The mean field assumption enters because we assume that the focal species interacts in an unbiased way with all other species, in proportion to their frequencies. This assumption results in the following generalization of Eq. 17 for typical fluctuation sizes of any of the species, j :

$$\Delta_j = \langle (p_j(\tau) - p_{j,0})^2 \rangle$$

$$= \tau \frac{p_{j,0}(1 - p_{j,0})}{K} (\sigma_j^2 (1 - p_{j,0}) + \frac{p_{j,0}}{1 - p_{j,0}} \sum_{i \neq j} \sigma_i^2 p_{i,0}) \quad [18]$$

shown in the main text as Eq. 1. I.e. we generalize σ_2^2 with the weighted sum over all (nonfocal) species: $\frac{\sum_{i \neq j} p_{i,0} \sigma_i^2}{\sum_{i \neq j} p_{i,0}}$.

Tests of mean field theory. Our mean field theory is an approximation. We test the accuracy of this approximation by simulating populations using our SLMCM described above, and then probing the output of the SLMCM, with the goal of assessing how accurately the theory predicts fluctuation sizes. To test our theory for fluctuation sizes in multispecies communities, explained in the main text and derived in *Mean Field Model for Fluctuations*, we now generate communities of $S = 10$ species, with 2-stage matrix population models. For any number of stages, the stochastic mean field model reduces to a single parameter (σ_j^2) for each species, and so the essence of this comparison is the same for any number of stages greater than 1. Hence, for simplicity, we selected 2 life stages (rather than eight, as in our empirical data below). We chose $\lambda = 1.1$ and imposed the constraint appropriate for competitor species derived above in Eq. 13, so that no species experiences a net, effective deterministic growth rate.

The key life history parameter emerging from the age-structured population dynamics is effective population sizes, N_y , or equivalently the year-to-year variance, $\sigma^2 = \frac{K}{N_y}$. This can be determined by competing each species against an identical copy of itself in the SLMCM, and then identifying the average time across replicates taken for one of the two copies to outcompete the other. This is analogous to the fixation time for an allele in population genetics competing against a different allele with the same fitness, though starting here at an initial frequency of 0.5. The relationship between N_y and the mean time to exclusion of one of the two species is given by the following formula:

$$\langle t(p) \rangle_{\text{neutral}} = 2N_y \left((1 - p) \log \left[\frac{1}{1 - p} \right] + p \log \left[\frac{1}{p} \right] \right), \quad [19]$$

as shown in e.g. ref. 106, so that

$$\langle t(1/2) \rangle_{\text{neutral}} = 2 \log [2] N_y. \quad [20]$$

The values of N_y (and hence σ^2) were determined for each synthetic species in our model by simulating 500 runs of competition of the species with a neutral

copy of itself, and taking the average of time taken for either copy to reach monodominance across these runs. Effective population size varied by around a factor of 40 from smallest to largest, with σ_i^2 correspondingly varying from 1.8 to 78. The specific matrices used to generate this range of values, and hence Fig. 1, are detailed in our [SI Appendix, section 2](#).

Using this synthetic community of ten species, we generated fluctuation sizes $\Delta_j = (p_j(\tau) - p_{j,0})^2$, for all species at every time point. In order to reduce these data to a form where we can plot each individual species' typical fluctuation size as a function of only its own initial frequency, we picked one focal species at a time, labeled j , chose its initial frequency, $p_{j,0}$, randomly from zero to 1, and then set each of the remaining species to have equal frequency $= \frac{1-p_{j,0}}{(S-1)}$. With this choice of initial conditions, Δ_j reduces to:

$$\Delta_j = \langle (p_j(\tau) - p_{j,0})^2 \rangle = \tau \frac{p_{j,0}(1-p_{j,0})}{K} \left(\sigma_j^2(1-p_{j,0}) + \frac{p_{j,0}}{S-1} \sum_{i \neq j} \sigma_i^2 \right) \quad [21]$$

In Fig. 1, we plot this theoretical prediction against observed, binned fluctuation sizes, for the species from our community with largest effective population size, and with $\tau = 5$ y.

Inference of Barro Colorado Island Matrix Population Models. To parameterize the life history strategies of the species modeled in this study, we combined data from two censuses: one of all individuals above 10 mm DBH at 5-y intervals (45) and one annual census focused on smaller individuals below that size threshold (46). Here, we will describe our methods for filtering species and data to obtain our 90 focal species and 8 size classes as well as the process of turning census observations into models. [SI Appendix, section 7](#) details a series of analyses demonstrating the accuracy of these life history estimates by comparing entries and N_y values of our 90 MPMs to matrices sampled from subsets of the available data.

Species selection. There are 277 species sampled from 1981 to 2015 which had an individual observed at DBH above 10 mm in all of the 8 censuses. We set a minimum threshold of 200 observed individuals in each census leaving 101 species. This was to ensure enough observations were present to confidently estimate vital rates across multiple size classes. We then matched the remaining species to a dataset estimating their minimum reproductive diameters, the diameter at which a majority of individuals of a given species are seen to produce fruits ([SI Appendix, section 3](#)), leaving 98 species covered by both studies. Finally, we integrated the data on smaller individuals to infer recruitment. All 98 remaining species were covered in the seedling and sapling dataset, but 2 were removed because no individual was observed to remain below 10 mm DBH for a 5-y period and 6 were removed because no individual below 10 mm DBH was observed growing to a diameter above that threshold. Both of these would conflict with our model assumption that individuals reproduce into the smallest size class and progress into larger classes at 5-y intervals. The full list of species names for the remaining 90 species is in our [SI Appendix, section 3](#).

Transition rate and recruitment rate estimation. To track vital rates across each species' life stages, we first split species into size classes based on their observed DBH. The objective for this analysis was to keep size classes consistent across species and thus a uniform set of breaks in diameters was used across all species. Several methods and algorithms have been developed for sorting individuals into stage and age classes, balancing biological realism, statistical confidence in parameter estimation, and specific goals of simulating various demographic processes (63, 107). We instead use an aggregate approach for all 90 species, breaking size classes into seedlings/saplings (individuals under 10 mm DBH from the small individuals censuses), and then 7 size classes at intervals of 10^{-3x} in mm DBH, for $x = 1$ through 6 (the last stage having no upper bound). The choice of log scale was motivated here by the well-described power-law distribution of tree diameters across multiple forest datasets (108, 109). Using these size steps, only 6 species had a stage with fewer than 20 individuals across all censuses. For these sparse stages, individuals in that

stage were instead moved into the previous stage to preserve statistical power in transition and recruitment estimates. After sorting individuals into stages, survival and transition rates were calculated based on observations between censuses. As an example, if 1,000 stage 3 individuals of a given species were present in the first seven censuses, half of them remained in that stage in the next census, 250 were observed to be in stage 4, 50 were observed to be in stage 2 (note that since these are size classes, individuals can revert to the previous stage in a 5-y period, even though this may be relatively rare), and 200 died, then the third column of the resulting survival matrix would be the following vector:

$$(0 \quad .05 \quad .5 \quad .25 \quad 0 \quad 0 \quad 0 \quad 0) \quad [22]$$

The smaller individuals (below 10 mm DBH) were tracked in a separate census. This census was conducted annually in nonconsecutive years. To match these data to the censuses of larger stems, we chose 5-y steps from 2001, 2006, and 2011 to match to the large stem censuses in 2000, 2005, and 2010. The transition and survival rates of these smaller individuals were then calculated the same way, tracking movement between stages for individuals across each 5-y interval to create the first column of the survival matrix.

Allometric scaling theory predicts a relationship of the form $R = \alpha D^z$ between reproductive biomass R and basal stem diameter D (65). There is substantial variation among estimates for the value of the scaling exponent z , with many reporting a value near 2, some finding smaller exponents, and others finding values much higher (65, 66, 69). We proceeded with a scaling exponent of 2, though we demonstrate that our predictions are robust to this scaling by considering recruitment exponents of both $z = 1$ and $z = 4$. As reported in the main text and in [SI Appendix, section 4](#), the resulting effective population sizes are largely insensitive to this choice, and the three sets of effective population sizes are highly correlated (Pearson correlation of at least 0.96 across the three comparisons). For the purposes of this study, we assumed that species have stable long-term growth rates $\lambda = 1$ as otherwise we would see rapid, deterministic exclusion of species with lower growth rates. Fixing λ and recruitment scaling this way, and assuming all new recruits are in the first stage, we can then solve for a unique fecundity matrix F , representing recruitment, such that:

$$F + U = A \quad [23]$$

$$A * v^0 = \lambda v^0 = v^0 \quad [24]$$

Based on each species' minimum observed reproductive diameter, between 1 and 4 size classes were able to produce offspring. For species with multiple reproductive size classes, recruitment rates increase for larger size classes so that $R \sim D^z$, z being 2 in the focal case. As the largest size class is unbounded, and thus has no maximum or mean value, we use the minimum diameter of the class as D . As an example, if one species had 2 reproductive stages, and those stages were consecutive (e.g. size classes 3 and 4), the smallest diameter in an individual in each size class would differ by a factor of 10^{-3} . If an individual in the third size class produced 100 offspring on average, one in the fourth would then produce $100 * (10^{-3})^2 = 398$, or roughly four times as many.

Computing effective population sizes. We compute σ^2 (the inverse of per capita effective population size) for our 90 Barro Colorado Island species by competing each species against an identical copy of itself using the SLMCM described above. Running multiple replicates of this neutral competition allows us to estimate effective population size, N_y , similarly to the method described for our synthetic age-structured populations in *Tests of Mean Field Theory*. The expected time to monodominance in these SLMCM simulations is given by formula [20]. For each of the 90 focal species in this study we averaged the results of 1,000 trials to obtain the values of σ^2 that are used to generate predictions for fluctuation sizes in our stochastic mean field model.

Environmental Noise Model. While in this manuscript we focus on the potential for life history variation to explain observed fluctuations, alternative models have introduced models of environmental stochasticity to explain the approximately power law scaling for the dependence of fluctuation size Δ_j on species j 's initial frequency, p_j (20). These models outperform models where the dependence of Δ on p is assumed to be linear. The linear dependence

represents model of purely demographic noise, though without the general stage-structure we have introduced here. Our analysis here is slightly different, as we are considering a subset of 90 species for which we were able to fit matrix population models, and these species tend to be among the more abundant species in the plot. We fitted a power law dependence via least squares, considering $\log(\Delta_{j,t})$ as a function of $\log(p_{j,t})$ for all species, j across all available censuses, t . We found the best fit power law parameters were as follows:

$$\Delta_{j,t} = ap_{j,t}^b \quad [25]$$

with overall factor $a = 3.10^{-4}$ and exponent $b = 1.61$, consistent with the scaling exponents between 1.5 and 2 identified in previous work analyzing abundance fluctuations on the Barro Colorado Island plot (e.g., ref. 20, Fig. 3).

Model Comparison. To address how closely our life history model can explain empirical fluctuation sizes, we now compare the outcomes of our model with the power law scaling predictions of environmental stochasticity described just above. In Eq. 1, we computed a mean field approximation to the expected value of Δ_j as a function of the frequencies and life history strategies of all species present in a community. To make contact between the various models considered here, we consider the deviation of model predictions from observations:

$$\log \sum_{j,t} \left(\log \Delta_{j,t,\text{obs}} - \log \Delta_{j,t,\text{pred}} \right)^2, \quad [26]$$

where in the case of our life history model $\Delta_{j,t,\text{pred}}$ is the prediction given by Eq. 1, and in the case of the power law fit, $\Delta_{j,t,\text{pred}} = b \log p_{j,t} + \log a$. The single-stage demographic noise model, in which changes in vital rates across life stages are ignored, systematically underpredicts fluctuation sizes as abundances increase, as shown in Fig. 4. Our 8-stage model, with quadratic scaling for recruitment with diameter, predicted fluctuation sizes more accurately in 64.4% of cases against the simpler model. I.e. for the majority of observations, the life history prediction for fluctuation size is closer to the observed fluctuation size, using the metric defined in Eq. 26. In *SI Appendix, Fig. 2*, we show corresponding comparisons of theory and data for linear and quartic scaling of recruitment with diameter, and these alternate versions of our model also outperform the single-stage demographic noise model (in 65.7% and 60.6% of cases, respectively).

We also find that our model makes predictions closer to observations than the power law model for environmental noise in 42% of observations, giving a slight advantage to the power law model. On the other hand, we do not tune

any parameters to fit observed fluctuation sizes, while the power law model has two free parameters. Given the lack of tunable parameters in our model, we conclude that outperforming a fitted model in close to half of the observations indicates that life history is a key part of predicting abundance fluctuations on Barro Colorado Island. We note that for our alternative choices of recruitment scaling exponent (described in *Transition Rate and Recruitment Rate Estimation*), this figure was robust, with 42% of predictions closer to observations than the power law model when recruitment scaling was linear with diameter, and 39% of predictions closer to observations than the power law model when recruitment scaled as diameter to the fourth. In summary, incorporating life history variation across multiple life stages outperforms a demographic noise model with a single life stage, as shown visually in Fig. 4. Fitting environmental noise parameters does provide more accuracy, but at the expense of fitting additional unobserved parameters.

Data, Materials, and Software Availability. Data needed to generate the matrix population models shown in Fig. 2, and used to generate Figs. 3 and 4 are drawn from publicly available census data and seeding data. Synthetic matrices used to generate Fig. 1 are shown in *SI Appendix, section 2*. Code samples used to generate results and figures used R version 4.2.2 with reshape2 version 1.4.4, dplyr version 1.0.10, and tidy version 1.2.1., and are available in a Zenodo archive (110). The first two photographs in Fig. 2 were taken by Steven Paton, and the third by Carmen Galdames. All three are licensed as CC0 1.0 (public domain) and were sourced from the Smithsonian Tropical Research Institute Plant Field Observations collection. All other data are included in the manuscript and/or *SI Appendix*. Previously published data were used for this work (45, 46).

ACKNOWLEDGMENTS. J.P.O. acknowledges the NSF, Division of Environmental Biology (grant #2401124) the Simons Foundation (grant #376199). The Barro Colorado Island forest dynamics research project was made possible by NSF grants to S. P. Hubbell: DEB #0640386, DEB #0425651, DEB #0346488, DEB #0129874, DEB #00753102, DEB #9909347, DEB #9615226, DEB #9405933, DEB #9221033, DEB #9100058, DEB #8906869, DEB #8605042, DEB #8206992, DEB #7922197, support from the Center for Tropical Forest Science/Forest Global Earth Observatory, the Smithsonian Tropical Research Institute, the John D. and Catherine T. MacArthur Foundation, the Mellon Foundation, the Small World Institute Fund, and numerous private individuals, and through the hard work of over 100 people from 10 countries over the past three decades. The plot project is part the Forest Global Earth Observatory, a global network of large-scale demographic tree plots. We also acknowledge R. Foster and S.J. Wright, and the editor and four reviewers.

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